
Top-Level Pipelined Winograd Accelerator Core (winograd_top.sv)
d_tile [0 : 3][0 : 3]16 × INT16
16 × 16b
Input Transform Unitinput_transform_2d $V = B^T dB$Multiplierless 1D Col → 1D RowAddition/Subtraction Trees
16 × 18b
REG (288b)
[17:0]
Multiplier Arraymultiplier_array $M = U \odot V$16 Parallel Signed MultipliersINT18 × INT16 → INT34(Configurable 4, 8, 16 DSPs)
16 × 34b
REG (544b)
M [33:0]
Channel Accumulatorchannel_accumulator $M_{accum} = \sum_{c=1}^{C_{in}} U^{(c)} \odot V^{(c)}$16 Parallel 36-bit AccumulatorsPartial Sum Storage
16 × 36b
REG (36b)
M_{accum}
Output Transform Unitoutput_transform_2d $Y = A^T M_{accum} A$Multiplierless 1D Col → 1D RowAdd/Sub Trees → $Shift(>>>6)$Saturation Logic to INT16
4 × 16b
y_tile [0 : 1][0 : 1]4 × INT16 (2 × 2 Output)out_valid
u_tile [0 : 3][0 : 3]16 × INT16 ($U = GgG^T$)
16 × 16b
REG (16b)
U [15:0]
Control Inputsstart, first_ch, last_ch, cin_count
first_ch_d2, last_ch_d2
Winograd FSM Controller (winograd_controller)States: IDLE → PROCESS → WAIT_DRAIN → DONEHandshakes: in_ready, first_channel, last_channel, tile_done